Math 521

Homework 4

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Problem 1. *Prove Cauchy's inequality with $\epsilon > 0$: If $a, b \in \mathbb{R}$ then:*

$$ab \leqslant \frac{\epsilon a^2}{2} + \frac{b^2}{2\epsilon}.$$

Proof. Since $a, b \in \mathbb{R}$, we have:

$$0 \leqslant (a-b)^2.$$

Expanding the right-hand side and isolating the ab term yields:

$$ab \leqslant \frac{a^2}{2} + \frac{b^2}{2}.$$

Now let $\epsilon > 0$. Since this holds true for all $a', b' \in \mathbb{R}$, take $a' = \sqrt{\epsilon}a$ and $b' = \frac{b}{\sqrt{\epsilon}}$. Then:

$$\begin{aligned} ab &= \sqrt{\epsilon}a \cdot \frac{b}{\sqrt{\epsilon}} \\ &\leqslant \frac{(\sqrt{\epsilon}a)^2}{2} + \frac{\left(\frac{b}{\sqrt{\epsilon}}\right)^2}{2} \\ &= \frac{\epsilon a^2}{2} + \frac{b^2}{2\epsilon}. \end{aligned}$$

□

Problem 2. *Let $U \subset \mathbb{R}^n$ be open, bounded, and ∂U is C^1. Let $u \in C^2(\overline{U})$ with $u = 0$ on ∂U. Prove the following interpolation inequality:*

$$\int_U |Du|^2\,dx \leqslant \frac{\epsilon}{2}\int_U (\Delta u)^2\,dx + \frac{1}{2\epsilon}\int_U u^2\,dx$$

Proof. From the previous problem, take $a = u(x)$ and $b = \Delta u(x)$. Then:

$$-u(x)\Delta u(x) \leqslant \frac{\epsilon(\Delta u(x))^2}{2} + \frac{(-u(x))^2}{2\epsilon}.$$

Integrating both sides over U gives:

$$-\int_U u\Delta u\,dx \leqslant \frac{\epsilon}{2}\int_U (\Delta u)^2\,dx + \frac{1}{2\epsilon}\int_U u^2\,dx$$

Since $u = 0$ on ∂U, we can rewrite the left-hand side of the inequality as:

$$-\int_U u\Delta u\,dx + \int_{\partial U} \frac{\partial u}{\partial \nu} u\,dS \leqslant \frac{\epsilon}{2}\int_U (\Delta u)^2\,dx + \frac{1}{2\epsilon}\int_U u^2\,dx.$$

Green's identity yields:

$$\int_U Du \cdot Du \, dx \leqslant \frac{\epsilon}{2} \int_U (\Delta u)^2 \, dx + \frac{1}{2\epsilon} \int_U u^2 \, dx.$$

Rewriting $Du \cdot Du = |Du|^2$ gives the desired inequality. ◻

Problem 3. *Assuming* f *aproaches zero rapidly enough as* $x \to \pm\infty$*, find a function* u *which satisfies the ODE problem:*

$$u''(x) - u(x) = f(x).$$

Your solution should be of the form:

$$u(x) = \int_{-\infty}^{\infty} K(x,t) f(x-t) dt.$$

Proof. After taking the Fourier transform of both sides, our ODE becomes:

$$(iy)^2 \hat{u} - \hat{u} = \hat{f}.$$

Solving for $\hat{u}$ gives:

$$\hat{u}(y) = \frac{-1}{1+y^2} \hat{f}.$$

Define $h(y) = \frac{-1}{1+y^2}$. Taking the inverse Fourier transform of both sides yields:

$$\begin{aligned} u(x) &= \mathcal{F}^{-1}(h(y)\hat{f}(y)) \\ &= \left(\check{h} * f\right)(x) \\ &= \int_{-\infty}^{\infty} \check{h}(t) f(x-t) dt. \end{aligned}$$

We previously showed that $\mathcal{F}(e^{-a|x|}) = \frac{1}{\sqrt{2\pi}} \frac{2a}{a^2+y^2}$. Hence:

$$\check{h}(y) = \mathcal{F}^{-1}\left(\frac{-1}{1+y^2}\right) = -\frac{\sqrt{2\pi}}{2} e^{-|x|}.$$

Thus the solution to our ODE is:

$$u(x) = -\frac{\sqrt{2\pi}}{2} \int_{-\infty}^{\infty} e^{-|t|} f(x-t) dt.$$ ◻

Problem 4. *For* $x \in \mathbb{R}$*, define* $v(x) := e^{-\frac{1}{2}x^2}$*. Show that* $\hat{v}(y) = e^{-\frac{1}{2}y^2}$.

Proof. ◻

Problem 5. *Let* $a > 0$ *and* $f \in L^1(\mathbb{R})$*. Define* $g(x) := f(ax)$*. Show that:*

$$\hat{g}(y) = \frac{1}{a}\hat{f}\left(\frac{y}{a}\right).$$

Proof. $\square$